Day 4 Mission: Discover How Your Computer Organizes Files

Today is not about learning Linux commands.

Today is about answering a question that most developers never deeply think about:

> How does a computer know where everything is stored?

> Without using File Explorer & any GUI App, you need to use only Terminal.

Every application, project, photo, video, operating system file, and configuration on your machine exists somewhere.

Your mission is to explore how that organization works, take screenshots & document them.

Definition of Done

By the end of today, you should be able to explain:

1. What is a file?
2. What is a directory?
3. What is the difference between them?
4. How does navigation work?
5. What is a path?
6. Why are paths required?
7. How does the operating system organize information?
8. How would you explain the filesystem to a new developer?

Most importantly:

You should understand these through exploration, not memorization.

Part 1 - Understand Your Current Location

Open the terminal.

Before typing anything, ask yourself:

> If I open a terminal right now, where am I?

The terminal is always operating from a specific location in the filesystem.

Your first challenge is to discover:

- How to identify your current location
- What information the terminal provides
- How the operating system represents locations

Document what you find.

Part 2 - Explore the Filesystem

Your computer already contains thousands of files and directories.

Your next challenge:

Discover how to view the contents of your current location.

Questions to investigate:

- What files exist here?
- What directories exist here?
- How does the terminal distinguish between them?
- What information can you learn just by observing?

Do not rush.

Look carefully.

Notice patterns.

Part 3 - Navigate Intentionally

Now begin moving through the filesystem.

Your goal is not to memorize commands.

Your goal is to understand:

> How does the operating system represent movement between locations?

Explore:

- Moving into directories
- Moving back
- Moving multiple levels
- Returning to known locations

While doing this, constantly ask:

> Where am I now?

> How does the operating system know that?

Part 4 - Create Your Own Structure

Create a directory called:

StarterWave

Inside it create:

- Projects
- Notes
- Experiments

Requirements:

- Terminal only
- No File Explorer
- No dragging and dropping

Once completed:

Inspect the structure.

Verify that everything exists exactly where you intended.

Part 5 - Investigate Files

Directories organize information.

Files contain information.

Create multiple files.

Experiment with:

- Creating files
- Renaming files
- Moving files
- Copying files
- Deleting files

Observe:

- What changes?
- What stays the same?
- How does the filesystem react?

Record your findings.

Part 6 - Understand Paths

This is one of the most important concepts today.

Every file has a location.

Every directory has a location.

The operating system identifies them through paths.

Your challenge:

Investigate:

- Absolute paths
- Relative paths

Then answer:

> Why are paths necessary?

> How would the operating system locate a file without them?

Do not search for textbook definitions.

Build your own understanding.

Part 7 - Break Something and Recover

Create a test directory.

Inside it:

- Create files
- Move files
- Rename files
- Delete files

Then intentionally create confusion.

Move things around.

Lose track of a file.

Find it again.

The goal is to experience how navigation and organization actually work.

Real understanding appears when you recover from confusion.
